

Parameterizing Waveform Uncertainty with Bilby for PE Marginalization

01-19-2024

What is the effect of waveform uncertainty on the posterior of BNS PE?

1. Generate waveform uncertainty data
 - a. Amplitude uncertainty and residual phase uncertainty, δA and $\delta\phi_R$
 - b. Parameterize the data so that large datasets can be efficiently produced
2. Create a calibration marginalization file with the parameterized waveform uncertainty data
 - a. Function in bilby to marginalize over calibration uncertainty
 - b. We give it waveform uncertainty as if it was calibration uncertainty
3. Reweight the posterior of a BNS PE run with the file
 - a. PE run without the waveform uncertainty
 - b. Create a data_dump.pickle file with the calibration uncertainty file
 - c. Reweight the posterior with the data_dump.pickle file
 - i. This process is very buggy and has yet to work

Generating Waveform Uncertainties

- Generate two waveforms with different models
 - IMRPhenomPv2_NRTidalv2 and SEOBNRv4T_surrogate
 - Waveforms are a series of complex strain values
- Calculate uncertainties with the formulas below:

$$s_1 = a_1 + b_1 i$$

$$s_2 = a_2 + b_2 i$$

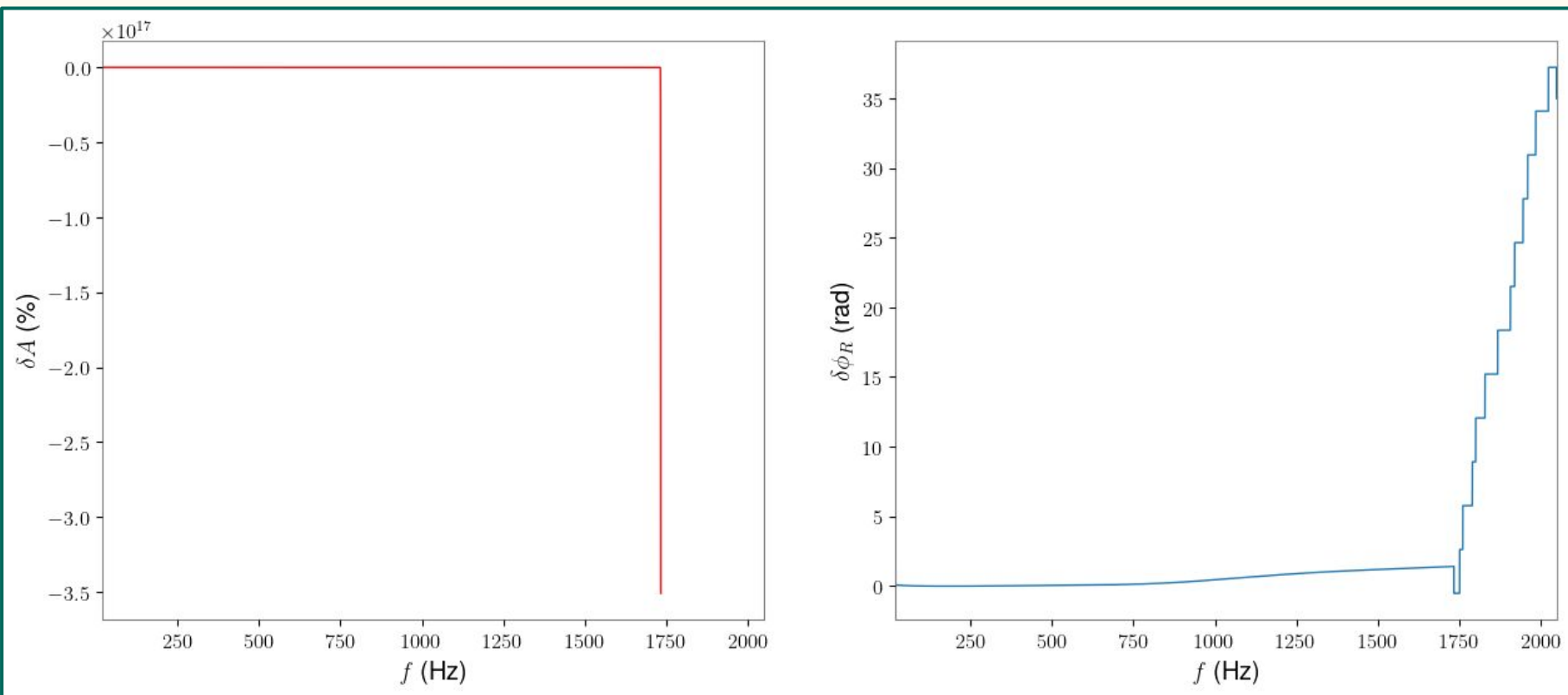
$$\delta A = \frac{|s_1| - |s_2|}{|s_1|}$$

$$\delta\phi = \arctan\left(\frac{b_1}{a_1}\right) - \arctan\left(\frac{b_2}{a_2}\right)$$

$\delta\phi$ has an overall phase and time shift

- We need to remove this overall phase
- Fit $\delta\phi$ to a line weighted by the PSD data and subtract the line
- Results in $\delta\phi_R$

Waveform Uncertainty Plots



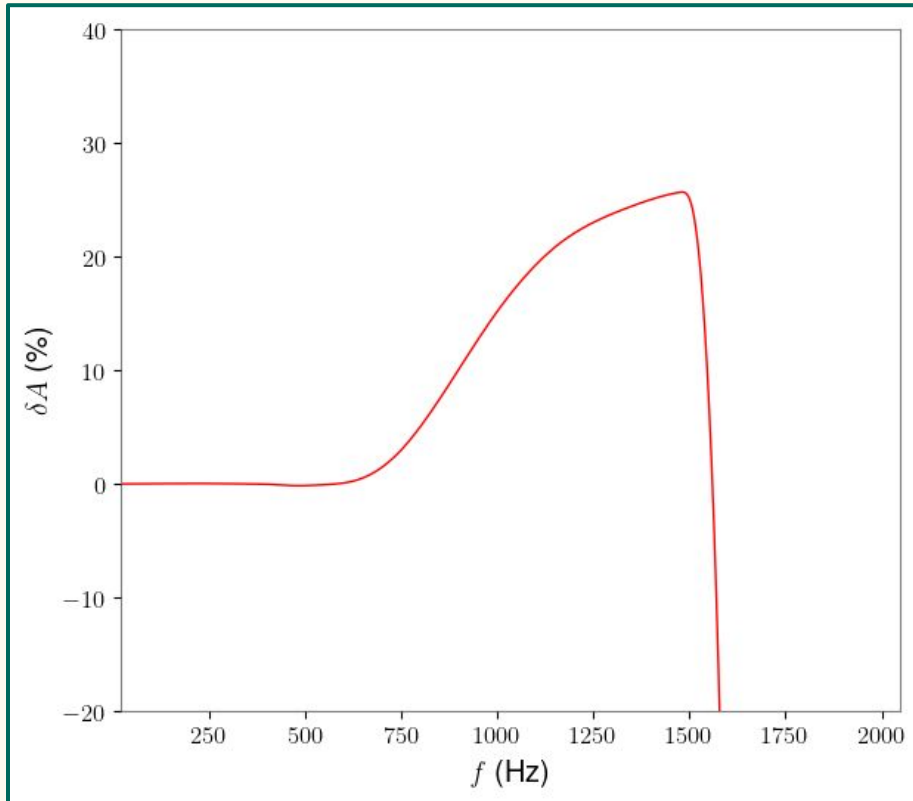
Tidal effects end BNS mergers early!

- Waveform models often disagree when the waveform should end
 - IMRPhenomPv2_NRTidalv2 always ends before SEOBNRv4T_surrogate
 - The following limit explains the result:

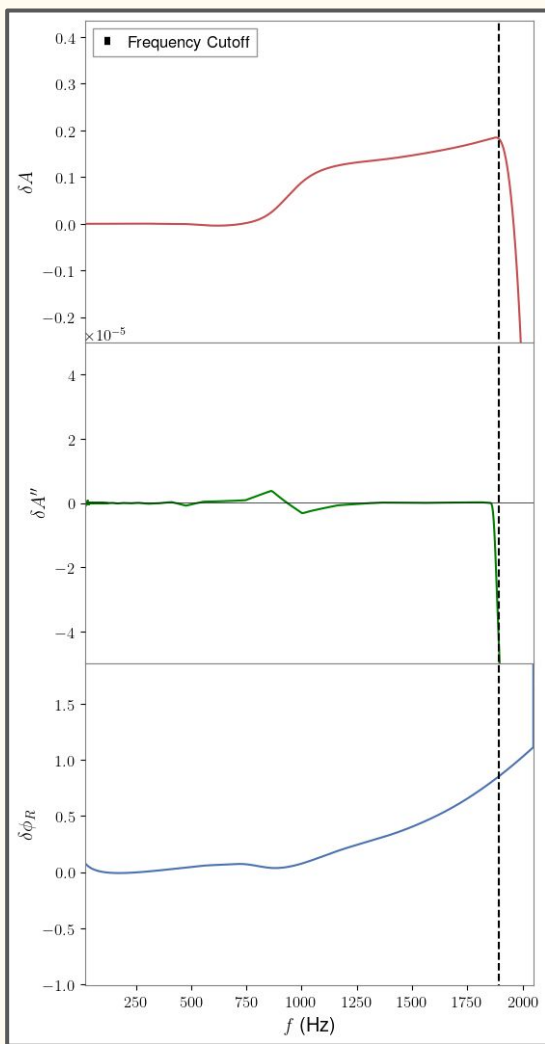
$$\lim_{s_1 \rightarrow 0} \left(\frac{|s_1| - |s_2|}{|s_1|} \right) = -\infty$$

- Stop the waveform before this happens
 - Frequency grid must stay in tact
 - $\delta\phi_R$ needs to end at the same point

Zoom in:



- Abrupt corner where $|s_1|$ starts to rapidly decay
- Automatically find this point
- Cut off the rest of the waveform



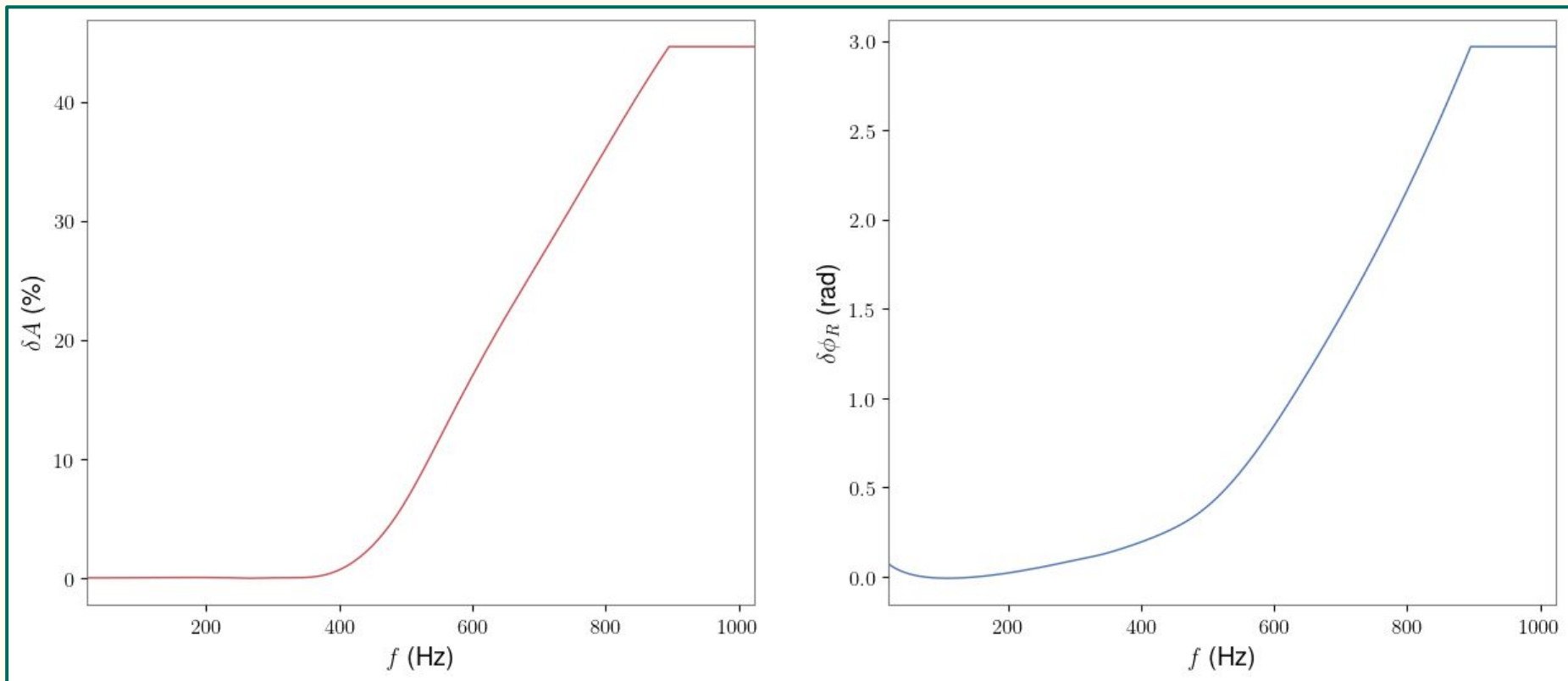
Use second derivative of δA

- Consistently finds the corner
- Marks a cut-off frequency to end the waveform before the discontinuity

What do we do after this cut-off frequency?

- After a waveform has ended, we just need to fill in values
- Hold uncertainty value constant for all frequencies

Final Waveform Uncertainties



Parameterization: Curve-Fitting

- Parameterize data to reduce the size of the sample sets
 - 520,000 points to 17 parameters
 - All original points recoverable
 - Set an error boundary
- Curve fit with Chebyshev Polynomials
 - Fit the data up to the cut-off frequency
 - Hold constant afterwards

δA and $\delta \phi_R$ as Functions of Frequency

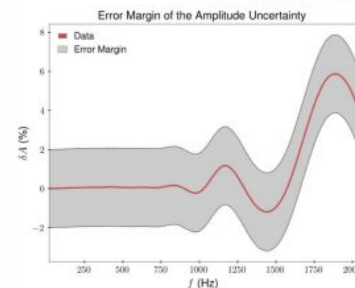
$$\begin{aligned}
 T_0(x) &= 1 \\
 T_1(x) &= x \\
 T_2(x) &= 2x^2 - 1 \\
 T_3(x) &= 4x^3 - 3x \\
 T_4(x) &= 8x^4 - 8x^2 + 1 \\
 T_5(x) &= 16x^5 - 20x^3 + 5x \\
 T_6(x) &= 32x^6 - 48x^4 + 18x^2 - 1 \\
 T_7(x) &= 64x^7 - 112x^5 + 56x^3 - 7x \\
 T_8(x) &= 128x^8 - 256x^6 + 160x^4 - 32x^2 + 1 \\
 T_9(x) &= 256x^9 - 576x^7 + 432x^5 - 120x^3 + 9x
 \end{aligned}$$

$$\delta A(f) \sim \sum_{i=0}^{N-1} a_i T_i(f) \quad \delta \phi_R(f) \sim \sum_{i=0}^{N-1} b_i T_i(f)$$

$[a_0, \dots, a_{N-1}]$ and $[b_0, \dots, b_{N-1}]$ would be the sets of parameters that represent the waveform uncertainties, where N is the number of parameters wanted.

Figure 1. The first 10 Chebyshev polynomials². The curve-fits are expressed as a series of these polynomials, each multiplied by a constant, a_i and b_i . These N constants would be the parameters that define the parameterized waveform uncertainties.

How good do the fits need to be?



The LIGO detectors, as with all equipment, have some uncertainties in their measurement. During the O3a observation period, these uncertainties were³:

2% Uncertainty in Amplitude
2° Uncertainty in Phase

Using these detector uncertainties, we were able to incorporate error margins into the waveform uncertainties, as shown in Figure 2.

If the curve fit stayed within this margin, it was accepted as a proper representation of the data. This allowed our number of fit parameters to be decreased substantially, as the accuracy was not as important.

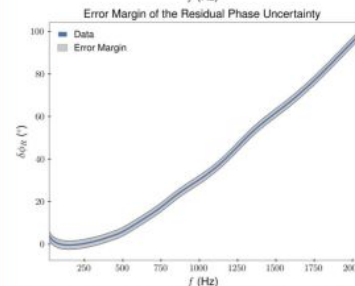


Figure 2. Example of each waveform uncertainty, δA and $\delta \phi_R$. Using the LIGO detector uncertainties, an error margin was defined, shown as the gray-shaded area. This margin for $\delta \phi_R$ is far stricter than that of δA , which forced the number of fit parameters to be exactly 10 rather than below it. δA itself can be fit in 6 parameters.

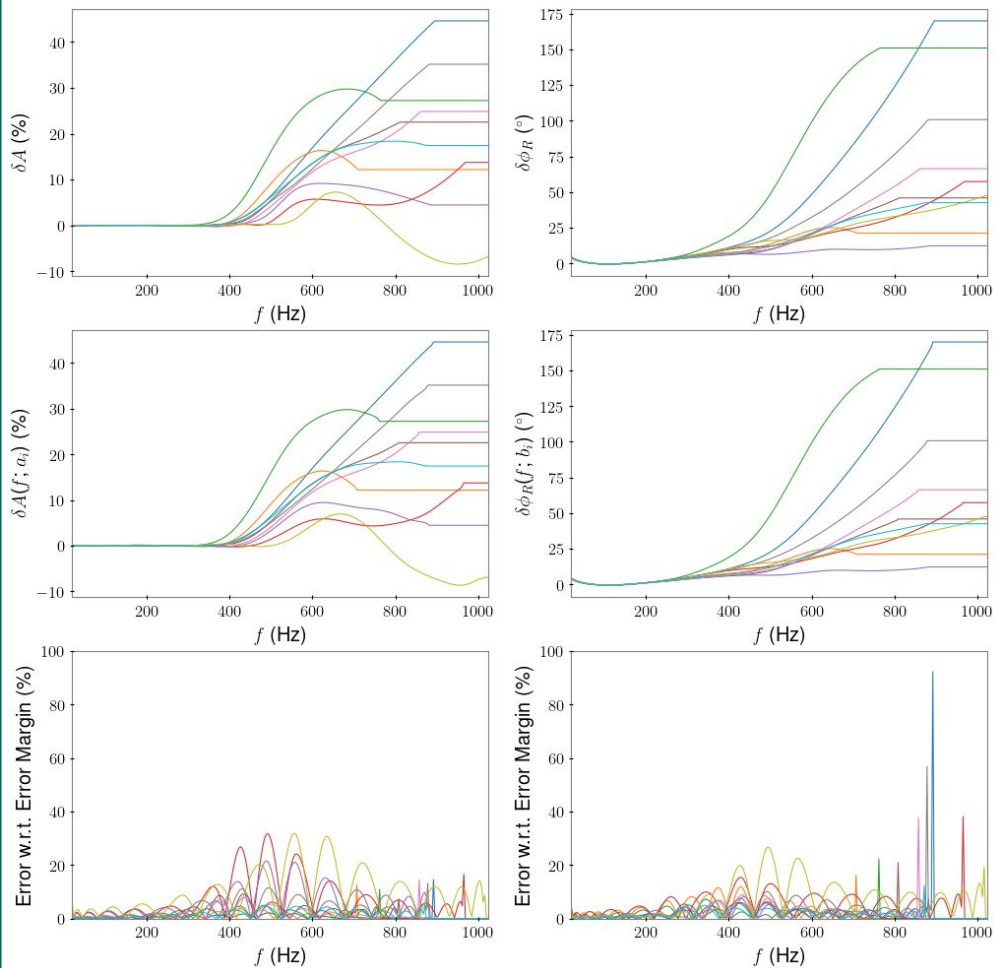
How the Data is Stored

Index	Frequency Grid	dA Parameters	dphi_resid Parameters	Final Index	Final dA Value	Final dphi_resid Value	System Parameters
0 47435	[20.0, 20.078948998975747, 20.158209645173454, ...]	[0.003238708807230789, -0.0003594191845902964, ...]	[0.14806884586444902, -0.004681710390600095, 2...	966	0.446106	2.967511	{'chirp_mass': 1.1976289821336776, 'mass_ratio'...
1 45904	[20.0, 20.078948998975747, 20.158209645173454, ...]	[-0.007999250408231628, 0.0009924048022438943, ...]	[0.08392541103413662, 0.0019306201854621709, -...	907	0.121851	0.368403	{'chirp_mass': 1.197716965454953, 'mass_ratio'...
2 35997	[20.0, 20.078948998975747, 20.158209645173454, ...]	[0.009285309669539464, -0.001032722436013697, ...]	[0.12058333456182829, -0.0010554592123680144, ...]	926	0.272422	2.635393	{'chirp_mass': 1.1977539491772493, 'mass_ratio'...
3 22192	[20.0, 20.078948998975747, 20.158209645173454, ...]	[0.022130676233658476, -0.002301517489100409, ...]	[0.18641681313169645, -0.009149092434087357, 0...	986	0.137377	1.001106	{'chirp_mass': 1.197515217891582, 'mass_ratio'...
4 19304	[20.0, 20.078948998975747, 20.158209645173454, ...]	[0.020741618191124453, -0.0021802008001914988, ...]	[0.13154020880282802, -0.0033863533329196423, ...]	963	0.044613	0.214545	{'chirp_mass': 1.1974684251736316, 'mass_ratio'...
5 25515	[20.0, 20.078948998975747, 20.158209645173454, ...]	[0.007830363569710248, -0.0008339913199121121, ...]	[0.12837635142687615, -0.0038293652161985473, ...]	941	0.225389	0.80063	{'chirp_mass': 1.197609587207121, 'mass_ratio'...
6 16197	[20.0, 20.078948998975747, 20.158209645173454, ...]	[0.009363557433991455, -0.0010230732553312186, ...]	[0.12696736924359175, -0.003847254570645282, 1...	956	0.248549	1.158485	{'chirp_mass': 1.19756424490524, 'mass_ratio'...
7 49068	[20.0, 20.078948998975747, 20.158209645173454, ...]	[0.003379993903749587, -0.0004005335083089504, ...]	[0.159955304338051, -0.006461062609114336, 6.4...	962	0.351511	1.758867	{'chirp_mass': 1.1975601555476183, 'mass_ratio'...

Data Recovery from Matrix

```
def recovery(identity,data):  
  
    if str(identity) == 'dA':  
        parameterized_curve = np.polynomial.chebyshev.chebval(data[1][0:data[4]],data[2])  
        post_waveform_uncertainties = np.copy(data[1])  
        post_waveform_uncertainties[data[4]-2:] = data[5]  
  
    elif str(identity) == 'dphi_resid':  
        parameterized_curve = np.polynomial.chebyshev.chebval(data[1][0:data[4]],data[3])  
        post_waveform_uncertainties = np.copy(data[1])  
        post_waveform_uncertainties[data[4]-2:] = data[6]  
  
    else:  
        raise Exception('Identity of the uncertainty must be "dA" or "dphi_resid".')  
  
    return np.concatenate((parameterized_curve[:-2],post_waveform_uncertainties[data[4]-2:]))
```

Parameterized Waveform Uncertainties

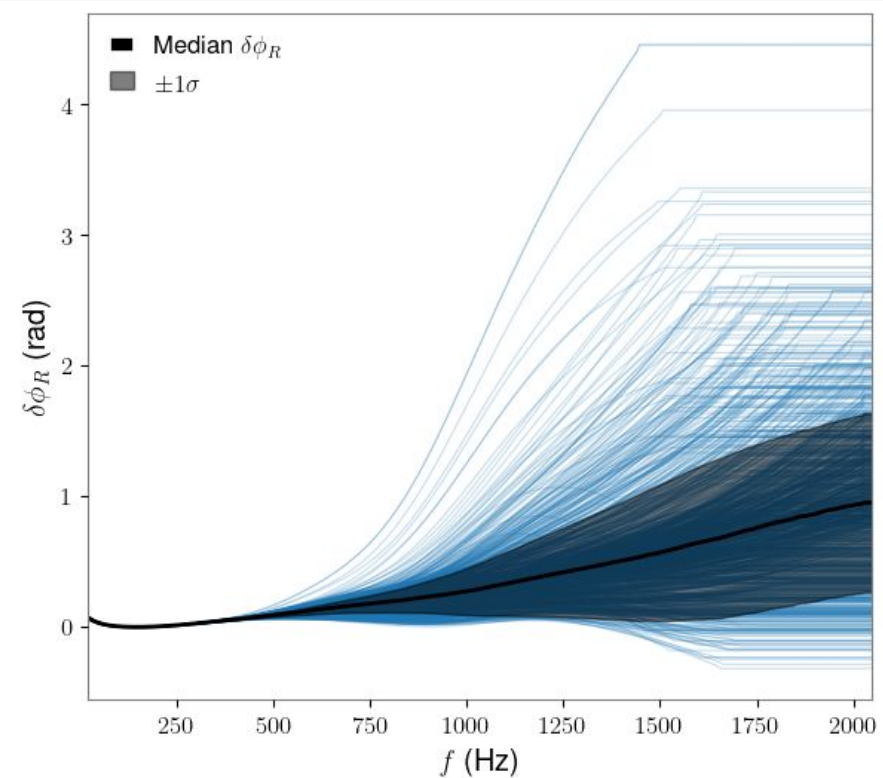
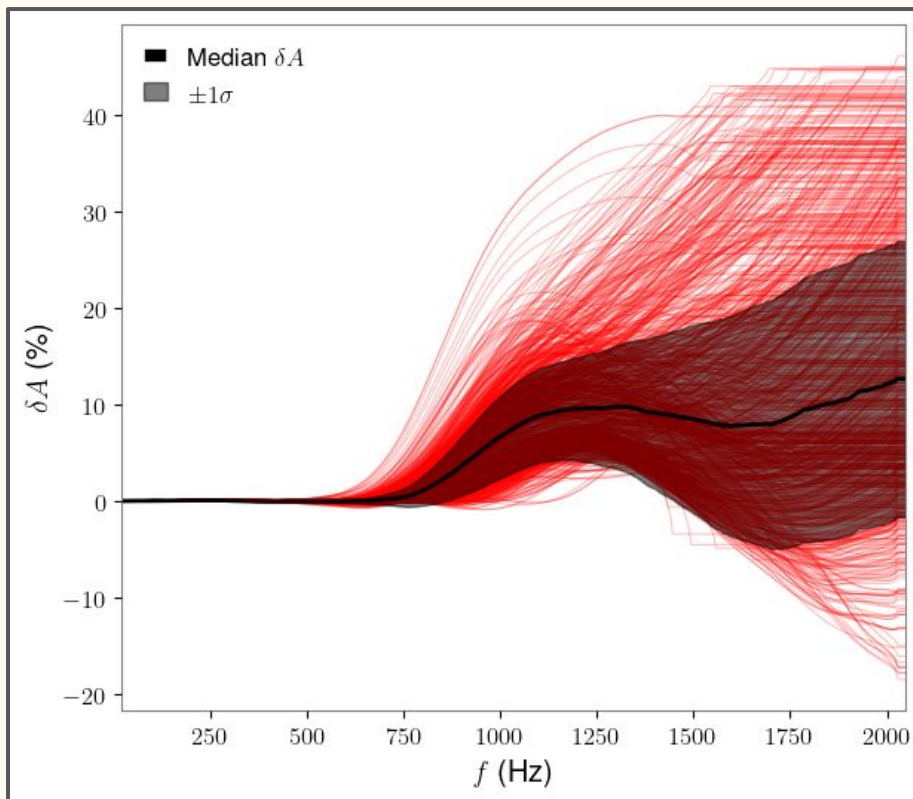


- Waveform Uncertainties
- Parameterized Uncertainties
- Error w.r.t. Error Margin
 - 100% error = margin border

Creating a Calibration Marginalization File

- 7 column matrix
 - Frequency (Hz)
 - Median Magnitude (%)
 - Median δA
 - Median Phase (rad)
 - Median $\delta\phi_R$
 - 1 Sigma Magnitude
 - 1 Sigma Phase
 - +1 Sigma Magnitude
 - +1 Sigma Phase
- Variable Frequency Grid

1	#	Frequency	Median Mag	Phase (Rad)	-1 Sigma Mag	-1 Sigma Phase	+1 Sigma Mag	+1 Sigma Phase
2		+2.0000000e+01	+1.0022497e+00	-9.3557530e-03	+9.6976742e-01	-3.9478890e-02	+1.0347320e+00	+2.0767392e-02
3		+2.0125000e+01	+1.0020021e+00	-9.1363100e-03	+9.6939211e-01	-3.9370224e-02	+1.0346121e+00	+2.1097604e-02
4		+2.0250000e+01	+1.0017619e+00	-8.9161290e-03	+9.6902525e-01	-3.9261625e-02	+1.0344985e+00	+2.1429367e-02
5		+2.0375000e+01	+1.0015289e+00	-8.6953410e-03	+9.6866673e-01	-3.9153210e-02	+1.0343911e+00	+2.1762528e-02
6		+2.0500000e+01	+1.0013031e+00	-8.4740740e-03	+9.6831642e-01	-3.9045085e-02	+1.0342898e+00	+2.2096937e-02
7		+2.0625000e+01	+1.0010842e+00	-8.2524490e-03	+9.6797421e-01	-3.8937349e-02	+1.0341943e+00	+2.2432451e-02
8		+2.0750000e+01	+1.0008722e+00	-8.0305850e-03	+9.6763996e-01	-3.8830097e-02	+1.0341045e+00	+2.2768927e-02
9		+2.0875000e+01	+1.0006670e+00	-7.8085930e-03	+9.6731358e-01	-3.8723420e-02	+1.0340203e+00	+2.3106234e-02
10		+2.1000000e+01	+1.0004683e+00	-7.5865830e-03	+9.6699495e-01	-3.8617402e-02	+1.0339416e+00	+2.3444236e-02
11		+2.1125000e+01	+1.0002761e+00	-7.3646570e-03	+9.6668395e-01	-3.8512125e-02	+1.0338683e+00	+2.3782811e-02
12		+2.1250000e+01	+1.0000903e+00	-7.1429170e-03	+9.6638049e-01	-3.8407664e-02	+1.0338001e+00	+2.4121830e-02
13		+2.1375000e+01	+9.9991069e-01	-6.9214580e-03	+9.6608388e-01	-3.8305513e-02	+1.0337375e+00	+2.4462597e-02
14		+2.1500000e+01	+9.9973724e-01	-6.7003720e-03	+9.6579331e-01	-3.8207800e-02	+1.0336812e+00	+2.4807056e-02
15		+2.1625000e+01	+9.9956980e-01	-6.4797490e-03	+9.6551000e-01	-3.8111274e-02	+1.0336296e+00	+2.5151776e-02
16		+2.1750000e+01	+9.9940828e-01	-6.2596740e-03	+9.6523387e-01	-3.8015993e-02	+1.0335827e+00	+2.5496645e-02
17		+2.1875000e+01	+9.9925255e-01	-6.0402290e-03	+9.6496482e-01	-3.7922015e-02	+1.0335403e+00	+2.5841557e-02
18		+2.2000000e+01	+9.9910252e-01	-5.8214930e-03	+9.6470274e-01	-3.7829393e-02	+1.0335023e+00	+2.6186407e-02
19		+2.2125000e+01	+9.9895808e-01	-5.6035420e-03	+9.6444754e-01	-3.7738175e-02	+1.0334686e+00	+2.6531091e-02
20		+2.2250000e+01	+9.9881912e-01	-5.3864500e-03	+9.6419912e-01	-3.7648410e-02	+1.0334391e+00	+2.6875510e-02
21		+2.2375000e+01	+9.9868556e-01	-5.1702860e-03	+9.6395739e-01	-3.7560138e-02	+1.0334137e+00	+2.7219566e-02
22		+2.2500000e+01	+9.9855728e-01	-4.9551190e-03	+9.6372226e-01	-3.7473403e-02	+1.0333923e+00	+2.7563165e-02
23		+2.2625000e+01	+9.9843419e-01	-4.7410140e-03	+9.6349363e-01	-3.7388239e-02	+1.0333747e+00	+2.7906211e-02
24		+2.2750000e+01	+9.9831620e-01	-4.5280330e-03	+9.6327142e-01	-3.7304682e-02	+1.0333610e+00	+2.8248616e-02
25		+2.2875000e+01	+9.9820321e-01	-4.3162380e-03	+9.6305553e-01	-3.7222764e-02	+1.0333509e+00	+2.8590288e-02



$$\Delta A_{\mu}(f; \theta_i) = \frac{|\mu_{IMR}(f; \theta_i)| - |\mu_{EOB}(f; \theta_i)|}{|\mu_{IMR}(f; \theta_i)|}$$

$$\Delta \Phi_{\mu}(f; \theta_i) = \tan^{-1}\left(\frac{\text{Im}[\mu_{IMR}(f; \theta_i)]}{\text{Re}[\mu_{IMR}(f; \theta_i)]}\right) - \tan^{-1}\left(\frac{\text{Im}[\mu_{EOB}(f; \theta_i)]}{\text{Re}[\mu_{EOB}(f; \theta_i)]}\right) - \phi_{0i} - 2\pi f t_{0i}$$

$$\delta \Phi_{\mu}(f) = \sqrt{\frac{\sum_{i=1}^N \Delta \Phi_{\mu}(f; \theta_i) - \overline{\Delta \Phi_{\mu}(f)}}{N}}$$

$$\delta A_{\mu}(f) = \sqrt{\frac{\sum_{i=1}^N \Delta A(f; \theta_i) - \overline{\Delta A_{\mu}(f)}}{N}}$$

$$\Delta A_{\mu}(f; \theta) \approx \Delta A_T(f; a) = \sum_{i=0}^{N-1} a_i T_i(f), \Delta \Phi_{\mu}(f; \theta) \approx \Delta \Phi_T(f; b) = \sum_{i=0}^{N-1} b_i T_i(f)$$

$$\mathcal{L}(h(f_j)|\theta, \alpha, \beta) = \frac{1}{2\pi P(f_j)} \exp(-2\Delta f \frac{[h(f_j) - \mu(f_j; \theta)(1 + \Delta A_{\delta}(f_j; \{f_n, \alpha_n\})) \exp[i\Delta \Phi_{\delta}(f_j; \{f_n, \beta_n\})]]^2}{P(f_j)})$$